Tien Pham

(Pham Canh An Tien)

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Web: Portfolio

Education

- **University of Manchester** — **Manchester, UK.**
  PhD student in Computer Science — October 2023 - September 2026
  Interactive Language Explanation and Visualization for Explainable Robotics
  Advisor: Angelo Cangelosi, Ph.D.
  Research Interest: Explainable Deep Reinforcement Learning, VLMs, Robotics
- **University of Texas at Arlington** — **Arlington, TX, U.S.**
  Master of Science in Computer Science — January 2021 - June 2023
  GPA: 4.00/4.00
  Thesis: Context-Aware Gaze-Based Interface and Smart Wheelchair
  Advisor: Manfred Huber, Ph.D.
- **University of Technology, Vietnam National University** — **Ho Chi Minh City, Vietnam**
  Bachelor of Engineering in Mechatronics (PFIEV Program) — July 2014 - July 2019
  Addendum to the Engineering Diploma of The Program of Training Engineers of Excellence in Vietnam
  GPA: 8.21/10.00
  Thesis: Design of Udulating Fin Applying for Autonomous Underwater Vehicle *(Score: 9.54/10.00)*
  Advisor: Tan-Tien Nguyen, Ph.D.

Research Experience

- **Cognitive Robotics Lab** — **Manchester, UK**
  Marie Curie Early Career Researcher — October 2023 - September 2026
  - Study on Explainable Deep Reinforcement Learning, Vision Language Models (VLMs) for Generalizable, Explainable, Trustworthy, Transparent and Interpretable Robot.
  - *Skills:* Computer Vision, Vision Language Models (VLMs), Task And Motion Planning (TAMP), Explainable Reinforcement Learning, Attention Mechanism, Human Robot Interaction.
- **Learning and Adaptive Robotics (LEARN) Lab** — **Arlington, TX, U.S.**
  Graduate Research Assistant — September 2022 - June 2023
  - Developing the Context-Aware Interface for disabled to interact with the wheelchair hand-freely while implement autonomous control for safety assurance.
  - *Skills:* Computer Vision, ROS, C++, Head pose Estimation, Eye gazing estimation, Object Detection, Newton Gaussian Method, MATLAB
- **Wireless and Sensor Systems Lab** — **Arlington, TX, U.S.**
  Graduate Research Assistant — January 2021 - May 2022
  - Worked on wearable devices for Medical Application such as FaceSense, BioFace, Seizure Detection on Wearable Device
  - *Skills:* Signal Processing, Machine Learning, Python, Data Analysis, Convolutional Neural Network
- **Robert BOSCH Engineering & Business Solutions Vietnam** — **Ho Chi Minh City, Vietnam**
  Embedded Software Developer — September 2019 - December 2020
  - Designed and executed platform of AUTOSAR Architecture for ECUs in automotive domain.
  - Developed OSEK Network Management for integrating to AUTOSAR Architecture.
- **Vietnam National DCSELab** — **Ho Chi Minh City, Vietnam**
  Undergraduate Research Assistant — July 2016 - July 2019
  - Designed and developed the gymnotiform undulating fin module.
  - Developed the computational model simulating the process of undulating fin in CFD.
  - Modeled the forces, moment, and other external forces affecting to the swimming process of the fin.

Teaching Experience

- Graduate Teaching Assistant, COMP34212: *Cognitive Robotics*, 2024, 2025, 2026, UoM.
- Graduate Teaching Assistant, COMP23311: *Software Engineering*, Fall 2023, UoM.
- Graduate Teaching Assistant, CSE 5301: *Data Analysis and Modeling Techniques*, Fall 2022, Spring 2023, UTA.
- Graduate Teaching Assistant, CSE 6331-004: *Advanced Topics in Database Systems*, Summer 2022, UTA.
- Graduate Teaching Assistant, CSE 3320-001: *Operating Systems*, Summer 2021, UTA.

Selected Publications

- Khang Nguyen, An T Le, **Tien Pham**, Manfred Huber, Jan Peters, Minh Nhat Vu (2025). *"FlowMP: Learning Motion Fields for Robot Planning with Conditional Flow Matching"* in Proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025).
- **Tien Pham**, Angelo Cangelosi (2025). *"Pay Attention to What and Where? Interpretable Feature Extractor in Vision-based Deep Reinforcement Learning"* in Proceedings of The 38th International Joint Conference on Neural Networks (IJCNN 2025).
- Tuan Dang, Trung Tran, Khang Nguyen, **Tien Pham**, Nhat Pham, Tam Vu, and Phuc Nguyen (2022). *"IoTree: A Battery-free Wearable System with Biocompatible Sensors for Continuous Tree Health Monitoring"* in Proceedings of the 28th Annual International Conference on Mobile Computing and Networking (MobiCom 2022).
- Vimal Kakaraparthi, Qijia Shao, Charles Carver, **Tien Pham**, Nam Bui, Phuc Nguyen, Xia Zhou, and Tam Vu (2021). *"FaceSense: Sensing Face Touch with An Ear-worn System."* in Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (pp. 1-27).
- Yi Wu, Vimal Kakaraparthi, Zhang Li, **Tien Pham**, Jian Liu, Phuc Nguyen (2021). *"BioFace-3D: Continuous 3D Facial Reconstruction through Lightweight Single-Ear Biosensors"* in Proceedings of the 27th Annual International Conference on Mobile Computing and Networking (pp. 350-363).

Honors and Awards

- Travel grant for IJCNN 2025 conference.
- Marie Skłodowska-Curie Research Fellowship
- ACM SIGMOBILE Research Highlights 2022
- Scholarships in 10 consecutive semesters in the PFIEV program, from 2014 to 2019
- Excellent Student Award, 2018 & 2019

Research Visit

- **CNRS-LAAS Toulouse, France**. October, 2025
- **PAL Robotics, Spain**. January, 2025

Patents

- **Method and Apparatus for Continuous Plant Health Monitoring Using a Battery-free System with Biocompatible Implanted Sensors. (2022)**
- **Method and Apparatus for Continuous 3D Facial Reconstruction Through Lightweight Single-ear(2021).**

Skills & Competencies

- **Coding**: MATLAB, Python, C, C++, Javascript and SQL.
- **Competences**: Researcher, Undergrad and Graduate Teacher, Workshop Organizer, Conference Reviewer.
- **Skills**: Machine Learning, Robotics, Computer Vision, Signal Processing, Human Robot Interaction.
- **Frameworks**: Pytorch, Mujoco, Ultralytics, Sklearn, Tensorflow, Docker, ROS noetic, ROS humble.
- **Languages**: Vietnamese (Native), English (Fluent), and French (Intermediate).

References

Angelo Cangelosi, PhD
Professor of Machine Learning and Robotics
Department of Computer Science
The University of Manchester, Manchester, UK
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Federico Tavella, PhD
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Centre for Robotic Autonomy in Demanding and Long Lasting Environements (CRADLE)
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